

Supplement to *Relative Wealth Concerns, Network Aggregation, and Effective Risk Aversion*

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This supplement reports the simulation referred to in Section 5.4 of the paper. Notation follows the paper throughout: $\bar{\theta} = m / \sum_a c_a$ is the effective market coefficient, $\vec{c}$ solves $[\mathbb{D}_{\theta+\lambda} - \mathbb{D}_\lambda A]\vec{c} = \mathbf{1}$ with $\lambda_a = \lambda$ common, θ_a is net risk aversion and A the row-stochastic influence matrix.

1 How often does reversal occur?

Proposition 2 of the paper exhibits a network on which comparison eventually lowers $\bar{\theta}$, and Theorem 3 gives the condition $\vec{\pi}^\top \vec{\theta} < H(\theta)$ under which this happens. Neither says how common such networks are. That is a question about a distribution over networks rather than about the mechanism, and the answer depends entirely on which distribution is chosen. We report one.

Design

Fix $\lambda = 0.5$ and draw 20,000 independent economies at each of ten combinations of population size m and dispersion σ . Off-diagonal entries of A are uniform on $(0, 1)$ with zero diagonal, normalised by rows; net risk aversion is $\theta_a = \max\{1 + \sigma Z_a, 0.15\}$ with Z_a standard normal. The population size is held *fixed* within each row, so that comparison across rows identifies the effect of size and not of a changing ensemble. Seeds are $1000m + 10\sigma$, set in `scripts/sign_reversal.py`.

Two distinct events are recorded and should not be conflated:

$$\text{cumulative: } \bar{\theta}(0.5) < \bar{\theta}(0), \quad \text{marginal: } \bar{\theta}'(0.5) < 0. \quad (\text{S1})$$

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The derivative is computed analytically rather than by finite differences. Writing $M(\lambda) = \mathbb{D}_\theta + \lambda(\mathbb{I} - A)$ and $\vec{c} = M^{-1}\mathbf{1}$,

$$\vec{c}' = -M^{-1}(\mathbb{I} - A)\vec{c}, \quad \bar{\theta}' = -\frac{m \mathbf{1}^\top \vec{c}'}{(\mathbf{1}^\top \vec{c})^2}, \quad (\text{S2})$$

so each draw costs two linear solves with the same factorisation of M . Intervals are Wilson intervals at 95 per cent.

Results

m	σ	cumulative: $\bar{\theta}(0.5) < \bar{\theta}(0)$		marginal: $\bar{\theta}'(0.5) < 0$	
		frequency	95% interval	frequency	95% interval
3	0.4	0.0919	[0.0879, 0.0959]	0.0952	[0.0912, 0.0993]
3	0.8	0.0340	[0.0316, 0.0366]	0.0372	[0.0346, 0.0399]
6	0.4	0.0057	[0.0047, 0.0068]	0.0062	[0.0052, 0.0073]
6	0.8	0.0008	[0.0005, 0.0014]	0.0008	[0.0005, 0.0012]
10	0.4	0.0001	[0.0000, 0.0004]	0.0001	[0.0000, 0.0004]
10	0.8	0.0000	[0.0000, 0.0002]	0.0000	[0.0000, 0.0002]
20	0.4	0.0000	[0.0000, 0.0002]	0.0000	[0.0000, 0.0002]
20	0.8	0.0000	[0.0000, 0.0002]	0.0000	[0.0000, 0.0002]
60	0.4	0.0000	[0.0000, 0.0002]	0.0000	[0.0000, 0.0002]
60	0.8	0.0000	[0.0000, 0.0002]	0.0000	[0.0000, 0.0002]

Table S1: Frequency of negative cumulative and marginal effects at $\lambda = 0.5$, over 20,000 draws per row with Wilson intervals. All ten cells of the design are reported; entries of 0.0000 are exact zeros, no draw out of 20,000 producing the event, and 0.0002 is the upper endpoint of the 95% Wilson interval when no event is observed.

Under this design reversal is already below one per cent at $m = 6$ and indistinguishable from zero at $m \geq 10$. The two events move together, which is what one expects when $\bar{\theta}$ is close to monotone over $[0, 0.5]$, and the cumulative frequency is slightly the smaller of the two at every cell.

What the table does and does not show

The decay in m is a property of this ensemble. As m grows, the rows of A approach the uniform distribution, so the design converges to the mean-field case of Proposition 1 of the paper, in which the influence matrix is approximately doubly stochastic and Theorem 2 makes the marginal effect non-negative. The decay is therefore mixing, not size. Remark 5 of the paper makes this precise: replicating the network of Proposition 2 in disjoint blocks leaves $\bar{\theta}(\lambda)$ unchanged for every population size, verified up to $m = 3000$, and the reversal survives weak ties across blocks up to a mixing fraction of about $\epsilon = 0.1$. Real influence structures carry communities, hubs and correlation between characteristics and network position, none of which this ensemble contains.

The two dispersions are not a clean comparative static. Reversal is rarer at $\sigma = 0.8$ than at $\sigma = 0.4$, but the difference cannot be read as an effect of dispersion alone. The truncation at 0.15 binds for 14.4 per cent of draws at $\sigma = 0.8$ against 1.7 per cent at $\sigma = 0.4$, which changes the shape of the distribution of θ_a and raises $\mathbb{E}[\theta]$ from 1.00 to 1.06. We report both rows and draw no conclusion from their difference.

Nothing here bears on the theorems. Theorems 2 and 3 of the paper hold for every network satisfying Assumption 1. The table describes the measure of networks on which the two limits point in opposite directions under one particular sampling scheme, which is a question of prevalence and not of validity.

Replication

The table is produced by `scripts/sign_reversal.py` and the block construction of Remark 5 by `scripts/block_reversal.py`, in the repository

<https://github.com/renatovicente/price-of-risk-aggregation>

Both are self-contained, depend only on `numpy` and `scipy`, read no data and draw from fixed seeds; `./reproduce.sh` runs them together with the rest in about fifteen seconds.